An Explainable Hybrid Machine Learning Framework for Fake Job Posting Detection

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***Abstract*—Online recruitment systems have made it easier for people to gain access to job opportunities. The scale of these systems has also made it easier for misleading job advertisements to spread quickly. These misleading advertisements mislead job seekers by giving them false information about the job, exaggerating the offered salary, and stealing personal details. Most of the existing solutions are only text-feature-based and black-box models. This makes it difficult to trust and depend on the results of automated decision-making. To solve these problems, an interpretable machine learning framework is presented that combines textual features with credibility metadata. Various job-related text features are aggregated and analyzed using standard natural language processing techniques and TF-IDF vectorization. Other features are designed to identify patterns that go beyond text analysis. These are the grammar error ratio, the presence of suspicious keywords, salary transparency, and company-related features. The hybrid feature representation is evaluated using Random Forest, Support Vector Machine, XGBoost, LightGBM, and stacking ensemble models. The models are tested for validity using a stratified test. The experiment shows that the hybrid set of features enhances the ability to distinguish between genuine and fraudulent job ads. The stacking ensemble model has the best ROC-AUC value of 0.994 and the lowest false negatives. Although the stacking ensemble achieves the highest ROC–AUC, XGBoost provides a more balanced trade-off between recall, precision, computational efficiency, and interpretability, making it suitable for real-world deployment.**

***Index Terms*—Fake Job Posting Detection, TF-IDF, Explainable AI, SHAP, Hybrid Feature Engineering, Recruitment Fraud Detection, Text Classification**

I. Introduction

Online recruitment platforms have greatly improved the accessibility of job opportunities through the rapid and massive dissemination of job information. However, this accessibility has also led to the proliferation of deceptive job advertisements that take advantage of job seekers through deceptive salary information, sense of urgency, and unauthorized requests for personal and financial information [1], [5]. These practices not only lead to financial woes but also contribute to the deterioration of trust in online recruitment platforms.

The challenge of identifying deceptive job advertisements is complex since deceptive job advertisements are becoming increasingly indistinguishable from legitimate ones in terms of tone, format, and linguistic usage. The employment of professional language, deceptive organizational information, and effective messages makes it challenging to distinguish deceptive job advertisements from legitimate ones based on simple rules [2], [10]. Machine learning algorithms have improved the accuracy of identification; however, most of the current methods are text-based and focus on the problem as a text classification problem, a strategy also prevalent in related domains like fake news detection [19]. Very little work has been done on credibility-related features such as linguistic quality, salary transparency, and organizational features that could potentially identify inconsistencies in behavior that go beyond the surface level.

Equally importantly, many proposed approaches are black-box classifiers. Unlike accuracy-centric approaches, a real-world fraud detection system must be interpretable to enable trust-building and verification by users. The lack of transparent reasoning components, therefore, is a significant shortcoming of the current state of the art.

To overcome these shortcomings, an explainable machine learning approach is proposed that combines textual representations with credibility-aware metadata features. Linguistic quality scores, language patterns, and company transparency scores are combined with interpretable boosting models to enable accurate and transparent predictions for use in online recruitment systems.

II. Related Work

The early research work on the identification of counterfeit job postings gradually evolved from rule-based filtering to supervised text classification learning. Following the inadequacy of static keyword filtering approaches in dealing with the dynamic patterns of job posting frauds, the TF-IDF feature matrices along with Logistic Regression, Naïve Bayes, and Support Vector Machines classifiers [1], [3] were employed. The approaches primarily targeted document classification and performed better than rule-based filters. Tree-based ensemble learning models such as Random Forest, gradient boosting, and ensemble-based recruitment fraud detection frameworks [2], [6] further enhanced performance by identifying non-linear correlations among features in high-dimensional text feature spaces. Later, the idea of using deep learning methods emerged to model the contextual dependencies that go beyond the bag-of-words approach. Methods such as LSTM and CNN-LSTM hybrid models [7]–[9] attempted to directly learn the semantics of the sequences from the job descriptions. While these models were able to provide competitive results, they were heavily reliant on large amounts of training data and were still centered on the text itself, with very little incorporation of the credibility metadata.

Some attempts were made to use the features of metadata, such as corporate features or salary features [4], [5]. However, the overall use of linguistic quality features and credibility metadata as a whole has not been explored much. In particular, there is a gap in the research that attempts to evaluate the importance of these features in interpretable models. This is what gives rise to the idea of the present framework.

III. Dataset and Experimental Setup

A. Dataset Description and Partitioning

The dataset used in this study is the publicly available Fake Job Postings dataset obtained from Kaggle [16], including both legitimate and fraudulent postings. Each posting includes several text fields such as job title, company profile, description, requirements, and benefits, in addition to more formal information such as salary range, employment type, telecommuting availability, whether a company logo is available, screening questions, industry, and job function.

TABLE I
Dataset Description and Class Distribution

Description	Count
Total Job Postings	17,880
Legitimate Jobs (Class 0)	17,014
Fraudulent Jobs (Class 1)	866

The target variable, fraudulent, is a binary variable that indicates whether a posting is legitimate or fraudulent. The dataset includes 17,880 postings, of which 866 are fraudulent, resulting in an imbalanced class distribution. To ensure that the class distribution remains balanced during evaluation, the data are split into 80% for training and 20% for testing using stratified sampling. The detailed class distribution is shown in Table I.

B. Evaluation Metrics and Implementation Details

Performance is assessed using Accuracy, Precision, Recall, and ROC–AUC. While Accuracy provides an overall correctness measure,

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN},$$

Precision and Recall are used to better evaluate detection of fraudulent postings:

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}.$$

Recall is also very important in this regard, as a false negative occurs when a scam is missed, meaning a scam is not detected when it is actually present. We also provide ROC-AUC for measuring the separation of classes at different thresholds.

All experiments are conducted in Python with scikit-learn [15], XGBoost, and LightGBM. The same preprocessing and data split is applied for all models.

IV. Proposed Methodology and Architecture

A. System Overview

Our approach is straightforward: raw job postings are transformed into a hybrid feature space and then used for prediction by machine learning models. To maintain consistency, the same preprocessing and feature extraction pipeline is applied in both training and production environments. As illustrated in Fig. 1, the system first consolidates and cleans the text data, followed by feature extraction that captures both semantic and credibility aspects. These features are combined into a unified representation and passed to supervised classifiers, which generate predictions along with feature-based explanations for transparent decision-making.

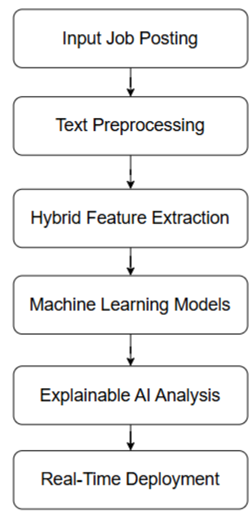

Fig. 1. System Architecture.

B. Text Preprocessing

All text fields are combined into a single document and preprocessed by converting to lowercase, removing URLs and special characters, normalizing spaces, lemmatizing words, and removing stopwords to reduce noise and ensure consistent feature representation.

C. Hybrid Feature Extraction

1) TF-IDF Textual Feature Representation: Next, we represent the text using TF-IDF weighting to emphasize words that are more important in this particular set of documents versus the overall corpus.

For a term t in document d:

$$TF(t, d) = \frac{f_{t,d}}{\sum_{t' \in d} f_{t',d}} \tag{1}$$

where $f_{t,d}$ denotes the occurrence count of t within d.

Corpus-level informativeness is captured through inverse document frequency:

$$IDF(t) = \log\left(\frac{N}{1 + |\{d \in D : t \in d\}|}\right) \tag{2}$$

where N represents the total number of documents and $|d \in D : t \in d|$ counts documents containing t.

The final weight is computed as:

$$TF\text{-}IDF(t,d) = TF(t,d) \times IDF(t) \quad (3)$$

Both unigram and bigram features are retained to identify standalone keywords as well as short phrases of context that often turn up in fraudulent postings. The sparse feature vector thus obtained captures the lexical patterns that are exploited by the downstream classifiers.

2) Suspicious Keyword Indicator: A predefined list of high-risk phrases such as "no experience" and "urgent hiring" is used to compute a frequency score. A binary indicator is also included to capture the presence of such keywords, allowing the model to learn their influence without relying on rule-based filtering.

3) Grammar Error Ratio (GER): For evaluating the quality of writing, a Grammar Error Ratio (GER) is calculated from the original aggregated job text before TF-IDF processing. For a document d, the text is broken into alphabetic word tokens through regular expression tokenization. The total number of extracted tokens is denoted by N_d, and the number of tokens not identified by a standard English dictionary through automated spell-checking is denoted by M_d.

The ratio is calculated as:

$$GER(d) = \frac{M_d}{N_d} \quad (4)$$

Spelling inconsistencies in fraudulent postings are generally higher, and hence this feature is a strong linguistic credibility marker.

4) Metadata and Credibility-Based Features: In addition to the text content, we have created credibility cues that are structured to identify inconsistencies at the presentation level. These credibility cues include structural statistics such as the length of the text and the number of words used, indicators for fishy keywords, grammar error ratio, and salary transparency is modeled using a binary indicator denoting the presence of salary information. We also examine company-related information such as the presence of a logo, presence of telecommuting claims, and the inclusion of screening questions. Taken together, these variables not only capture the text but also the structure and presentation of the job posting, which is a key contribution of this research.

5) Hybrid Feature Fusion: Textual and metadata representations are combined through horizontal concatenation. Let $\mathbf{X}_{\text{text}} \in \mathbb{R}^{n \times p}$ denote the TF–IDF matrix and $\mathbf{X}_{\text{meta}} \in \mathbb{R}^{n \times q}$ denote the engineered metadata features. The final feature matrix is constructed as:

$$\mathbf{X}_{\text{final}} = [\mathbf{X}_{\text{text}} \ \ \mathbf{X}_{\text{meta}}] \quad (5)$$

This unified representation allows the classifiers to jointly classify semantic content and credibility. The design principle of the proposed framework is the integration of sparse lexical features with structured metadata.

D. Machine Learning Models

The final hybrid feature matrix, $\mathbf{X}_{\text{final}}$, is then used with a range of supervised classifiers to test its ability to cope with linear and nonlinear learning algorithms. We utilize the Random Forest classifier, a linear SVM classifier, XGBoost, LightGBM, and a stacking ensemble classifier, all within the same training process, following prior machine learning–based classification approaches [17], [18]. Out of all of them, XGBoost is chosen for deployment based on its robustness and ability to perform SHAP interpretation.

1) Random Forest: Random Forest [13] is used as a bagging-based baseline to model non-linear feature interactions. The classifier is trained with 300 trees ($n_estimators = 300$) and no depth restriction. To improve minority-class exposure, controlled undersampling is applied to the training data before fitting, maintaining approximately a 3:1 ratio between legitimate and fraudulent samples. The other models are trained on the original imbalanced training data using class-weight balancing strategies. Parallel computation ($n_jobs = -1$) is used to improve efficiency.

2) Support Vector Machine: The linear SVM [14] is trained on the entire hybrid feature space. Due to the high-dimensional nature of the TF-IDF features, a linear kernel is employed. To handle the class imbalance problem, the linear SVM is trained with `class_weight="balanced"`. Because linear SVMs are not calibrated, a calibration step is performed using cross-validation to provide probabilistic outputs and facilitate ROC-AUC score calculation.

3) XGBoost: XGBoost [12] is trained with 300 boosting rounds with a maximum tree depth of 6 and a learning rate of 0.05. To reduce overfitting, both row subsampling and feature subsampling are set to 0.8. Since the dataset is imbalanced, the positive class is weighted using the ratio $\frac{N_0}{N_1}$. This setup provides a reasonable trade-off between recall and precision while remaining compatible with exact tree-based SHAP explanations.

4) LightGBM: LightGBM is trained with 400 boosting iterations and a learning rate of 0.05. Balanced class weights are applied to account for skewed class distribution. Feature subsampling and minimum child constraints are used to improve generalization in the sparse TF–IDF space. Its performance is comparable to XGBoost with slightly different precision–recall characteristics.

5) Stacking Ensemble: A stacking ensemble combines Random Forest, calibrated SVM, XGBoost, and LightGBM as base learners, similar to prior ensemble-based fraud detection approaches [6]. Their predictions are used as inputs to a Logistic Regression meta-classifier with balanced class weighting. The ensemble improves sensitivity to fraudulent postings but increases architectural complexity and slightly reduces precision, which influenced the decision to prioritize XGBoost for deployment.

V. Results and Discussion

The performance of the models is measured and tested on the held-out data, referred to as the test set, and the performance measures include accuracy, precision, recall, ROC-AUC, and absolute false negative counts. The goal is not only to achieve the highest possible accuracy but also to investigate how each classifier handles class imbalance and detects fraudulent postings.

The summary of the comparative results is shown in Table II. All models have achieved high classification accuracy above 97%, which implies that the proposed hybrid feature representation has good separability for legitimate and fraudulent samples.

Stable performance is demonstrated by the balanced nature of the tree-based boosting models. XGBoost demonstrates a recall of 0.84 with 28 false negatives, signifying the effectiveness of fraudulent posting detection. The precision is further enhanced as a result of this boosting algorithm. LightGBM shows a slightly improved accuracy of 99% compared to XGBoost, coupled with a precision of 0.96 and a minor increase in false negatives.

For the stacking ensemble, the ROC-AUC is maximized at 0.994, coupled with the lowest false negatives of 14. This indicates the sensitivity of this ensemble as a result of combining heterogeneous boundaries of decision-making. The precision is reduced to 0.60 as a result of a large false positive rate.

The Linear SVM and Random Forest classifiers serve as good baselines, but they have a slightly higher number of false negatives. These results indicate that the gradient boosting classifiers perform better for the sparse-dense hybrid feature space that we have constructed.

TABLE II
PERFORMANCE COMPARISON OF DIFFERENT MODELS

Model	Accuracy (%)	Precision	Recall	ROC–AUC	False Negatives
Random Forest	98.0	0.82	0.73	0.987	47
Support Vector Machine (SVM)	98.0	0.91	0.76	0.987	41
XGBoost	98.0	0.81	0.84	0.990	28
LightGBM	99.0	0.96	0.82	0.990	32
Stacking Ensemble	97.0	0.60	0.92	0.994	14

A more detailed view of the discriminative capacity can be obtained with the Receiver Operating Characteristic (ROC) analysis. In Fig. 2, the ROC for the XGBoost, LightGBM, and the stacking ensemble model can be found.

The stacking ensemble model has the highest value for the Area Under the Curve, which correlates with the lower FN value for the model. LightGBM and XGBoost are very close, with the ROC curves for the two models overlapping for most threshold values. This shows that the hybrid feature representation plays a larger part than the boosting algorithm used. While the stacking ensemble performs best with regard to ROC-AUC and false negatives, it compromises on precision, resulting in an increased number of false positives. This might not be ideal for real-world recruitment systems, where an increased number of false positives would lead to additional manual checks and affect user trust in the system. Also, the stacking architecture makes the inference process more complicated. XGBoost offers a better balance between recall and precision while maintaining computational efficiency and compatibility

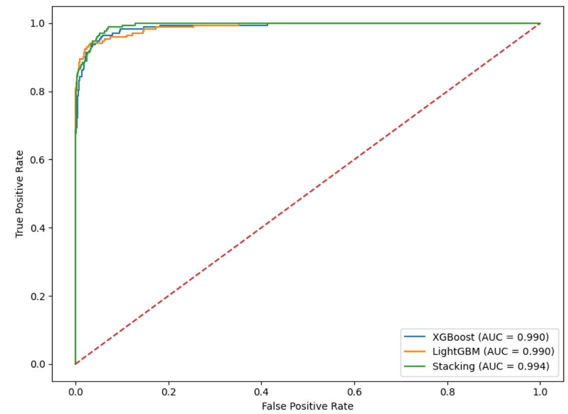

Fig. 2. ROC curves of XGBoost, LightGBM, and stacking ensemble models on the test dataset.

with SHAP-based interpretation. Hence, XGBoost is chosen for deployment and further explanation analysis.

The following section investigates the extent to which the performance trends are supported by feature attribution and not merely statistical discrimination.

VI. Explainable AI (XAI)

In the case of fraud detection for recruitment platforms, it is not enough for the model to be correct. The decision must be interpretable, particularly when the decision is wrong and has implications for real-life job opportunities. Explainability, therefore, is included in the process to evaluate the model's use of meaningful information for fraud detection, rather than relying on spurious correlations or dataset-specific artifacts.

The explanations for the model use SHapley Additive exPlanations (SHAP) [11], a technique widely validated for its trustworthiness in high-stakes domains such as healthcare [20]. For a prediction function $f(\mathbf{x})$, SHAP explains the prediction as:

$$f(\mathbf{x}) = \phi_0 + \sum_{i=1}^{d} \phi_i \tag{6}$$

where ϕ_0 is the expected model output, and ϕ_i stands for the contribution of the i-th feature. This form enables the direct inspection of the impact that the textual and credibility-based features have on the model's output.

A. Global Feature Analysis

The global behavior of the model is investigated through the SHAP summary plot presented in Fig. 3, in which the features are ordered according to their average absolute value of contribution to the final classification, creating a hierarchy of feature importance.

The feature importance hierarchy indicates that credibility-related metadata plays a significant role in the classification task within the trained model. Specifically, the presence of a company logo is among the most important features, indicating that transparency in companies is a strong indicator of authenticity. Furthermore, the Grammar Error Ratio is a feature with consistent contribution to the classification, indicating that the quality of language is a strong indicator of the probability of a fraudulent advertisement.

These findings verify the decision to incorporate credibility features in the classification model, as the model does not rely solely on suspicious keywords in the text, assigning a specific importance value to the quality of writing and the presence of a company logo, as presented in Fig. 3.

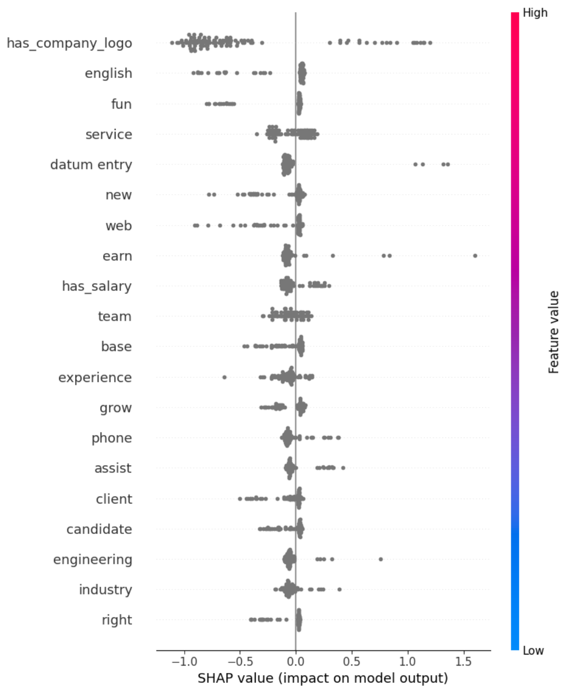

Fig. 3. Global SHAP feature importance highlighting the most influential textual and metadata features.

B. Local Prediction Analysis

While global rankings show general trends, predictions need instance-level reasoning for individuals. Fig. 4 shows a SHAP force plot for an example job posting.

The plot displays feature-level breakdowns of the prediction. Positive SHAP values move the model output to the fraud class, and negative values move it to the legitimate class. In the example, the high Grammar Error Ratio and lack of salary information and company branding positively influence the classification of the text as fraudulent.

This instance-level transparency is useful for manual verification by reviewers. Reviewers do not need to rely on probability values alone. This level of granularity is critical for operational use cases in recruitment scenarios.

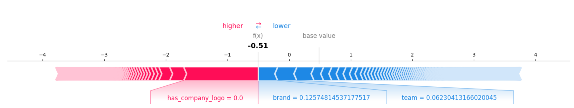

Fig. 4. Local SHAP explanation for an individual job posting prediction.

C. Interpretability-Driven Validation

The explainability analysis functions as validation of the proposed hybrid feature design. The prominence of company-related attributes in Fig. 3 confirms that structural credibility indicators influence model behavior. The measurable impact of Grammar Error Ratio demonstrates that linguistic quality operates as an authenticity signal rather than a decorative feature. Salary transparency variables further reinforce the role of metadata in fraud detection.

Many prior works report performance metrics without examining internal decision patterns. In contrast, the present framework uses interpretability to verify that the classifier relies on semantically and structurally meaningful indicators. The alignment between feature attributions and domain expectations strengthens confidence in deploying the system in real recruitment settings.

VII. Real-Time Deployment

A web interface is built using the Streamlit library to showcase the real-time nature of the fraud detection system. The deployed system loads the trained XGBoost model, along with the corresponding TF-IDF vectorizer and metadata configuration, to perform the actual predictions.

For each job posting, the model calculates a probability score for the occurrence of a fraudulent job posting. A threshold value can be set to convert the probability to a binary classification outcome, and the class label, along with the confidence level, is presented to the user.

In addition to this classification result, a dynamic instance-level SHAP explanation is created. Fig. 6 is a real-time SHAP waterfall plot that shows feature-level contributions to the prediction result. The features shown in red increase the fraud probability, while those shown in blue decrease it. This way, the model does not remain a black box but becomes transparent for each prediction made.

Fig. 5 shows the complete interface for the deployment of the model for both prediction and explanation, integrated into a single interface.

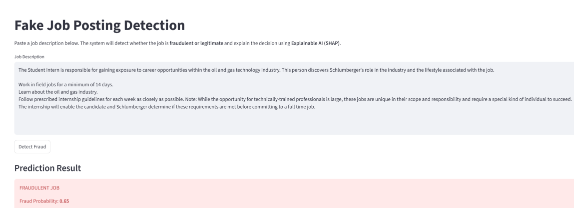

Fig. 5. Streamlit-based real-time interface for fake job posting detection.

VIII. Conclusion and Future Work

This paper proposes a hybrid framework for detecting fake job posts based on the fusion of TF-IDF text feature extraction and credibility-oriented metadata features. Besides text features, this proposed framework includes writing quality features such as Grammar Error Ratio, keyword frequencies of suspicious words, salary transparency features, and features of

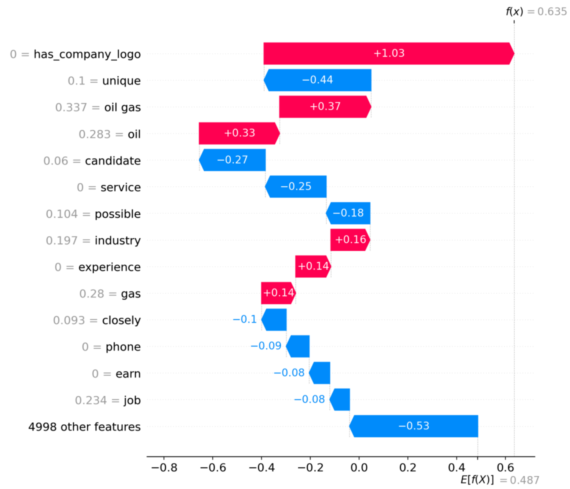

Fig. 6. SHAP explanation for individual instances generated during real-time prediction, explaining the contribution of features for fraud classification.

companies such as the presence of a logo. The experimental results demonstrate that the proposed framework can improve the performance of fraud detection by reducing false negatives compared to text feature extraction alone. Among all the proposed models, the stacking ensemble method has the best performance measured by ROC-AUC value and the lowest false negative rate; however, this is achieved at the expense of increased complexity and decreased precision. The XGBoost method is a balanced choice between recall rate, stability, efficiency, and compatibility of SHAP explanation. The interpretability of the proposed framework further confirms the importance of credibility features for fraud detection.

Future work will be aimed at extending the framework with transformer-based language representations for better contextual understanding while maintaining credibility features. Evaluation will be extended to multilingual and cross-platform datasets. Salary anomaly detection can be strengthened using market-aware benchmarks, and adaptive threshold strategies may further improve robustness under evolving fraud patterns.

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